Several simulation program systems are available for predicting the material properties of real composite materials, ranging from commercial finite element packages to specialized molecular dynamics and multiscale frameworks. These tools address different length scales and physical phenomena, and often are used in combination.

□ Commercial Finite Element and Multiphysics Software

These are the most widely accessible tools for simulating composite behavior at the macroscopic level.

* **Abaqus, Ansys, and LS-DYNA**: These commercial software packages are commonly used for modeling mechanical properties, thermal properties, and fabrication processes of composite materials . They can be used with user-defined subroutines (like UMAT or VUMAT) to implement custom material models .

* **DIGIMAT**: A specialized software for multi-scale material modeling, often used for mesoscale homogenization. It allows for the explicit modeling of fibers and matrix within a Representative Volume Element (RVE) to compute effective thermomechanical properties of laminates .

□ Molecular Dynamics (MD) and Coarse-Grained MD (CGMD)

For predicting properties governed by molecular-level interactions (like the interface in filled rubber), MD simulations are used.

* **LAMMPS**: A classical molecular dynamics simulation code. It has been used to study the mobility of polymer chains near filler interfaces, providing data that can be fed into larger-scale models .

* **Coarse-Grained MD (CGMD)**: This approach simplifies molecular models to extend simulation times and length scales. It has been used to model the behavior of uncrosslinked and crosslinked rubber, extracting parameters like the disentanglement time and diffusion coefficient that inform macroscopic constitutive models .

□ Multiscale and Deep Learning Frameworks

To bridge the gap between molecular details and macroscopic behavior, specialized multiscale frameworks are being developed.

* **Molecular-Informed Multiscale Framework**: A methodology that uses CGMD simulations to derive parameters (like relaxation time) and then maps them onto macroscopic viscoelastic constitutive models. This is implemented through a user-defined material subroutine (UMAT) in finite element software like Abaqus .

* **Deep Learning-Enhanced Multiscale Homogenization**: This approach uses deep neural networks (specifically Multilayer Perceptrons, MLPs) trained on data from high-fidelity multiscale homogenization models. It acts as a surrogate model to instantly predict effective properties, such as the thermal conductivity of composite aerogels, without running complex finite element simulations .

* **Direct FE² Method**: A framework that integrates two-scale models into a single finite element analysis, eliminating nested calculations. It can be implemented in commercial software like ABAQUS and is used to model the elastoplastic behavior of composite laminates .

□ Specific Applications and Models

Simulation frameworks are often tailored to specific material systems and properties.

* **Thermal Conductivity Models**: Simple models like the Wiener Parallel-Series model have been developed to predict the thermal conductivity of rubber-sand composites, offering a feasible solution for quantitatively evaluating porous geomaterials .

* **Aging and Degradation**: Finite element simulations can model corona-induced aging in silicone rubber by linking applied voltage to electron energy and bond-breaking reactions, which in turn affects water diffusion behavior . Coupled fluid-structure interaction (FSI) simulations in Abaqus are used to assess the degradation of elastomers under thermo-mechanical and chemical stresses in downhole conditions .

* **Sealing Performance**: An ultra-viscoelastic aging model implemented as a UMAT subroutine in ABAQUS has been used to numerically simulate the stress relaxation and waterproof performance of EPDM rubber seals in tunnels .

□ Summary of Key Considerations

The choice of simulation system depends on the **property to be predicted** and the **relevant length scale**. For macroscopic mechanical or thermal behavior, **Abaqus, Ansys, or DIGIMAT** are standard choices. To understand how molecular structure influences these macroscopic properties, **CGMD frameworks** coupled with **finite element methods** are used. Emerging **deep learning-based surrogate models** offer a computationally efficient path for predicting effective properties once trained on high-fidelity simulation data. The accuracy of these systems relies on the proper calibration and validation of the underlying material models and parameters.